2021 IEEE 6th International Conference on Intelligent Computing and Signal Processing (ICSP 2021)

Research on Visualization of Emergency Response To COVID-19

——Social Network Analysis Based on Microblog Data

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2021 6th International Conference on Intelligent Computing and Signal Processing (ICSP) | 978-1-6654-0413-6/20/$31.00 ©2021 IEEE | DOI: 10.1109/ICSP51882.2021.9409000

Abstract: **This paper is based on theories related to social network analysis, with Sina WeChat "emergency management" information dissemination as the research object, and web spider, Chinese word segmentation, Gephi and other social network analysis tools to carry out empirical study of emergency management structures, network centres and media at various phases during COVID-19. Also, in this paper, upon the empirical analysis of "emergency management" information dissemination, it has been found that the structure of the emergency management network gradually intensified during COVID-19.**

Key words: emergency preparedness plan, emergency supply, emergency measure, asymptomatic, social network analysis

I. INTRODUCTION

The outbreak of COVID-19 at the beginning of 2020, as a major public hygiene event, is the fastest spreading, the most extensive and challenging disease in China since its foundation. In this epidemic prevention and control, people all over China have made a lot of contributions as well as sacrifice. We are pulled together in times of trouble and thus learned a lot of experience. It should noted that most critical and important step in emergency management must be the capability to learn lessons from the crisis, upgrade risk assessment mechanisms and risk prevention and control capabilities, and identify and eliminate potential risks to further refine and optimize policies, thus enhancing "immune effectiveness" against new emergencies or crises. In this regard, in this paper, we would take COVID-19 outbreak beginning in late December 2019, when a small number of cases were detected in Wuhan, and becoming normal in mid-April 2020 as the subject to investigate the important measures and key directions of emergency management in the epidemic prevention and control process in China. In this paper, social network analysis method and CiteSpace V software are used to conduct quantitative statistical analysis on the emergency management process during the outbreak of COVID-19 in China. A visual knowledge map was drawn to illustrate the key measures and front-burner issues of emergency management during the epidemic. This provides some reference value for improving the traditional emergency management of public emergencies.

II. METHOD AND DATA

A. Research methods

Social network analysis has been proven as an important method to study the relationship between actors in quantitative sociology. This kind of research method regards the "relationship" between actors in social activities as the unit of analysis and deems structure as a pattern of relationships between actors. This method would be able to reflect the overall characteristics of network architecture, the position of the individual object in the network structure, and describe the extent to which the overall structure affects individual objects simultaneously.

Generally, integral structure of network is measured by using network density, average path length, clustering system and network centrality. Network density reflects the closeness of connections between nodes in a social network. In general, this can determine whether the network structure is loose or tight one [1]-[2]. The greater density of the network enables members to share implicit knowledge, less denss network enables members to share explicit knowledge. Average path length refers to the average distance between any two nodes in the network required to communicate. Small average path length indicates that the exchange of knowledge among members would be more convenient, conducive to the sharing and dissemination of knowledge. Clustering coefficient reflects the possibility of information exchange and cooperation between members according to the tightness of nodes. The average path length clustering coefficient and network node degree can be combined to determine whether it conforms to the power law distribution, thus it can further determine whether the network is a small world network or a scale-free network [3]-[5].

B. Data sources

Process consultation on emergency management during COVID-19;China's social medias include Sina Weibo, eWeChat, QQ, Baidu, etc. official medias include Ministry of Health of the Republic of China and Ministry of National Emergency Management etc. The author Selected Sina Weibo as the source of information and data selection, because it has greater influence, wide range of user group and a huge number of audience,

C. Data acquisition

In this paper, the following five date nodes are selected as the data collection acquisition. Phase I: December 27, 2019 solstice January 19, 2020: monitoring of unidentified pneumonia cases in Wuhan City, Hubei Province; China reported the epidemic for the first time; Phase II: January 20 to February 20: Initial containment of the epidemic: China took the key measure to stop the spread of the virus, a decisive quarantine started. Phase III: February 21 to March 17: The number of new cases in China has gradually dropped to single digits, and the CPC Central Committee has made a major decision to coordinate epidemic prevention and control, economic . Phase IV: March 18 to April 28: Decisive result was obtained in the "Battle" of Protecting Wuhan and the Battle of Protecting Hubei. With Wuhan as the main battlefield, The quarantine of Hubei was then released, COVID-19 patients in hospitals in Wuhan were cleared. Phase V: Since April 29: The epidemic prevention and control all over China has been norm, and the epidemic situation in China is now a sporadic pattern, with recurrent imported cases from abroad + asymptomatic cases.

The concrete operation mode in this paper: The Gooseeker web spider is used, keywords including "emergency preparedness plan, emergency supplies, emergency measure, asymptomatic, travelling by scan a QR code" are used to support the epidemic prevention and control, and retrieve micro blog and WeChat data (www.weibo.com). As of February 16, the last day of retrieval for this paper, a total of 29, 056, 079 microblogs were retrieved. Later, in this paper, 139, 348 microblogs in these ten periods were collected by Gooseeker; Using Python, this paper collected about 122,000 comments from these microblogs. After de-cleaning the data, they got about 87,000 comments. Finally, the word segmentation software of Gooseeker was used to segment the comments in these time periods, thus drawing the word cloud map.

III. Data Results and Discussion

A. *Overall characteristics of network*

Upon the data mining and social network analysis over these five phases, it has been found that:

Center node analysis: Phase I (27 December 2019 solstice 19 January 2020) Immediate response to outbreak of the epidemic: Central node: It includes time, service, manage properly, report, get united, department, health, issue convening, organization, request, etc., the centrality is not that prominent. It also reflects that the focus of emergency management was not prominent in the early phase of the epidemic outbreak. As shown in Figure 1. Phase II (January 20 to February 20), Initial containment of the epidemic, the center nodes include mobilization, strength and frontline, assistance, Party Committee, research, regions, etc., Centrality was neither obvious, however, it reflects the high difficulty of containment and prevention. As shown in Figure 2.

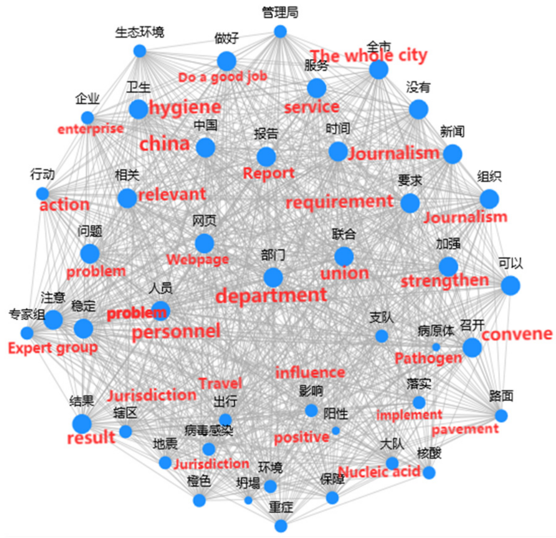

Figure 1 Phase I Structure Drawing

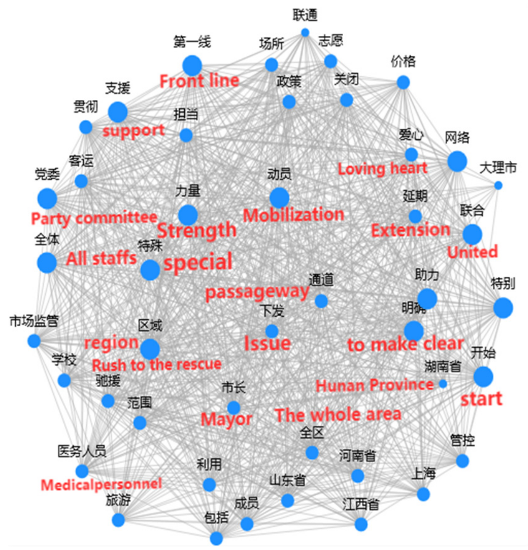

Figure 2 Phase II Structure Drawing

Key-node Analysis: Phase III focuses on Committee, followed by logistics and coordination. It reflects that the key to emergency response in the middle of an outbreak is through the Committee as a mediating point. At the same time, coordination and logistics become the key links in rescue as well as emergency service. The key node of Phase IV is Committee, followed by conditions and controls. It reflects that the critical point of emergency response in the middle and late phases of an outbreak is that the Committee is regarded as a mediator. It also meets the conditions and control measures. and it has become the key link of rescue work. As shown in the following figure.

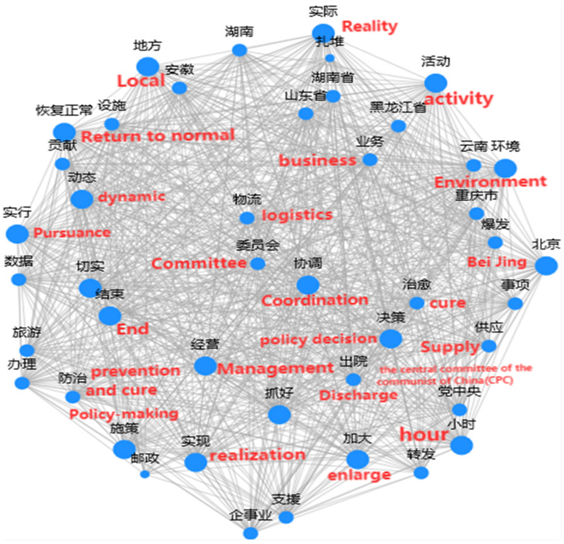

Figure 3 Phase III Structure Drawing

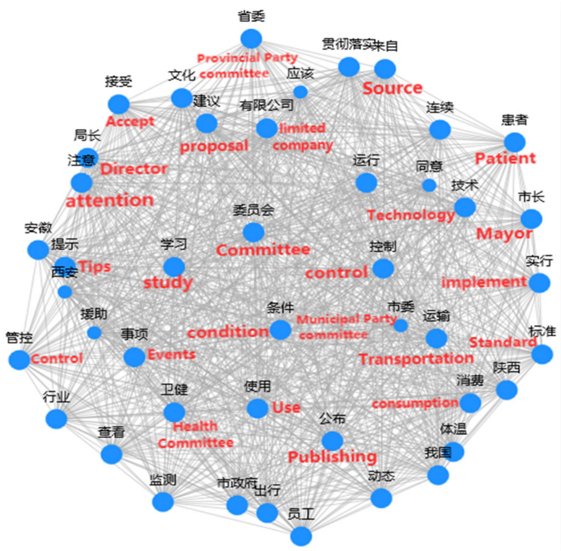

Figure 4 Phase IV Structure Drawing

Network density analysis: Network density is the commonly used indicator in social network analysis. In an undirected network, network density refers to the rhe ratio of the actual number of connections between network nodes to the maximum number of possible connections, with the expression formula of 2 l/ [n(n -1)]. Where, l is the number of connections existing in the real world, and n is the number of nodes. A large density means there are more connection lines between nodes in a network, and actors are more connected and information flows more smoothly; A small density means there are few connections between nodes and there are few connections between actors.. The density of the network can be calculated through Info/Network/General command in Pajek The minimum network density is 0.09 in the Phase V, which means the relationship of each department in the network is in a relatively loose state, and there is no relatively close connection. The maximum network density is 1.59 in the Phase IV, it means that the whole situation is a dynamic process. As events develop and time goes on, in the process of emergency management, the communication and collaboration between various departments changed from loose to tight and then to loose as the epidemic progressed, which is in line with the actual law of the development of things.

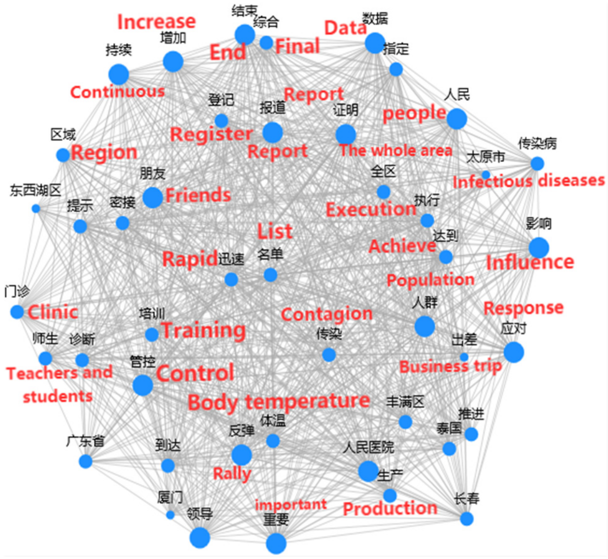

Figure 5 Phase V Structure Drawing

B. *Network centrality analysis*

Degree centrality reflects the relationship between knowledge nodes and so on in standard network, which reflects the degree of connection between the actors and the importance of the nodes, which would be able to measure the centrality of knowledge transfer. In terms of degree centrality, in the three phases before and after the COVID-19, the centrality of each phase is different. It is proved from the side that the phase key links of emergency management are also changing, and the top five concrete degree centers are shown in the following table:

TABLE I. Degree centrality of emergency response at all phases

Keywords	Absolute value	Relative value	Actual value	Keywords
Phase I	Expert group	6983.000	12.977	0.044
	Nucleic acid	6163.000	11.453	0.039
	Pathogen	5703.000	10.598	0.036
Phase II	Responsibility	558.000	0.034	0.034
	United	553.000	9.107	0.033
	Strength	552.000	9.091	0.033
Phase III	Policy-making	901.000	5.278	0.044
	CPCCC	894.000	5.237	0.044
	Decision-making	833.000	4.879	0.041
Phase IV	Municipal Party Committee	1743.000	8.140	0.051
	Municipal Government	1577.000	7.364	0.046
	Agree	1513.000	7.065	0.045
PHASEV	Achieve	882.000	7.083	0.039
	Rapid	859.000	6.898	0.038
	Rally	806.000	6.473	0.036

Seen from the table(s), we may reach a conclusion that the "expert group" in the phase I is the central link of emergency disposal, which confirms that the decision-making department of emergency disposal in the early phase is the opinions and suggestions of the expert group. "Responsibility" has been the central task of emergency response in the phase II. It reflects that the willpower of the whole country, the government and the people in the period of fighting the epidemic is an important factor in controlling the epidemic; "Policy-making" in the Phase III is the core work in the middle of the epidemic, reflecting that the specific core work in the emergency response phase is the implementation of prevention and control policies; The centrality of Phase IV is the highest departmental organizations, the keywords include "Municipal Party Committee" and "Municipal Government", "Provincial Party Committee" and "Consent", which reflect the authority leadership of emergency response; The centrality of Phase V is comparatively not obvious, "Rapid" "Rebound" "Control" and "Reach" jointly reflect the core words in the later phases of the epidemic.

IV. Conclusion

In this paper, social network analysis was adopted as the theoretical basis and "Sina Weibo" was selected as the sample data source. The emergency response process during COVID-19 was divided into five phases for data acquisition. Gephi group building, Pangu word segmentation tools were used to process and analyze keywords, which is a demonstration of different changes in different phases of emergency response during the COVID-19 epidemic. The global characteristics of network and network centrality are discussed and analyzed. The research result showed that: 1. In the prevention and control of COVID-19 epidemic, the central leadership in the emergency response process was the committee seen from the analysis of the center nodes and central nodes of the network; 2. When we analyze the changes in network density, the complexity and variability of emergency response can be fairly demonstrated. To be specific, as the epidemic progressed, network passwords went from tight to relatively loose. It reflects that the recovery of production and life after the epidemic is progressing in an orderly way. 3It can be concluded that the centrality of emergency response is changing and shifting with the change of anti-epidemic process from the analysis of centrality

Acknowledgments

The successful completion of this article is thanks to the support of the scientific Research Fund project of Yunnan Provincial Department of Education on the Improvement of Public Safety emergency Response Capacity by large numbers of technologies -- A Case Study of COVID-19 Epidemic "(Project No. 2021J0961).

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